

ProofBridge

Evidence-Based Talent Intelligence Platform for Academia & Industry

CORE THESIS

Transforming campus placements from resume-based guessing into **cryptographic, evidence-backed skill verification** and automated curriculum intelligence.

• LIVE ON LOCALHOST:3000

W3C DID SPEC COMPLIANT



The Crisis of Resume Inflation & Curriculum Blindspots

THE CORE PARADOX

Students claim skills they don't have. Recruiters spend **42 hours** filtering AI-generated CVs. Universities don't know what industry needs **until companies stop coming**.

01 — FOR STUDENTS

The Rejection Blackbox

78% of student resumes carry exaggerated or generic claims — "**React: Advanced**", "**Python: Expert**".

0 actionable feedback on rejection. Students never learn **which specific concept** disqualified them.

Ø Unverified MOOC certificates carry zero signal for an engineering manager.

02 — FOR RECRUITERS

Screening Paralysis

42h of engineering time burned per technical hire, manually parsing unverified resumes.

86% rise in AI-generated, keyword-stuffed CVs overwhelming ATS filters (Deloitte 2026).

CGPA and college prestige used as crude proxies — missing top-tier tier-2/3 builders entirely.

03 — FOR UNIVERSITIES & DEANS

Accreditation Blindspots

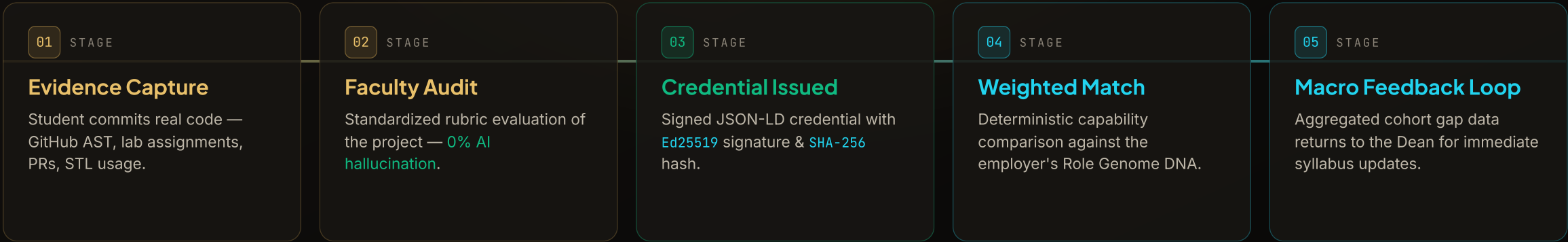
24–36 mo SOT curricula lag behind real production engineering stacks.

Months of manual paper audits for NAAC A+ and NBA Outcome-Based Education.

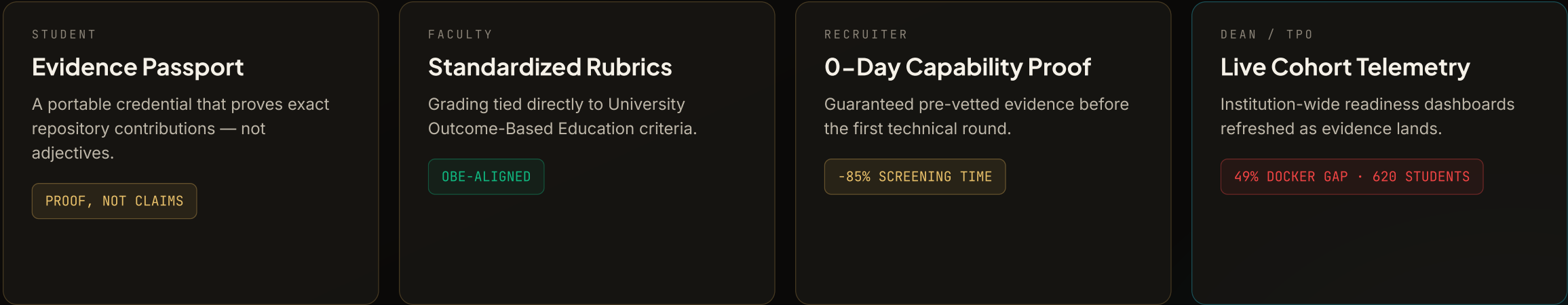
600 students, zero aggregate visibility into whether they truly know Docker or Spring Boot.

Solving the Four-Way Information Asymmetry

CONTINUOUS VERIFICATION PIPELINE



THE FOUR PRIMARY BENEFICIARIES



Beyond Job Listings — A True Talent Intelligence Layer

01 Skill Twin vs. Role Genome

Static bullet points are replaced with multi-dimensional capability vectors, resolved at four levels of mastery.

L1
Syntax

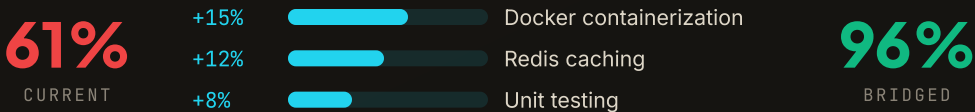
L2
Idiomatic /
Libraries

L3
Architecture

L4
Production Deploy

02 The “Bridge Me” Shortest-Path Engine

A real-time delta engine that computes the minimum work required to close a specific role gap.



03 Blind Vector Talent Radar

Pedigree, gender and brand bias are removed. Candidates rank purely on cryptographic proof vectors.

Candidate #A8217

Backend 88%

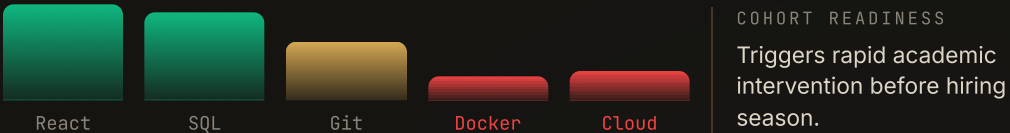
AST Depth: High

Security: Passed

Name / College: Hidden

04 Dean Curriculum Gap Radar

Corporate hiring requirements are scanned and mapped against university-wide readiness in real time.



Evidence Over Claims: What Real Code Looks Like

GITHUB AST PARSER · STATIC ANALYSIS

CANDIDATE A — THE CLAIMED RESUME

“Advanced C++ & Backend Developer”



INSPECTION LOG

- ✗ Only 1 C++ repository found
- ✗ 88% template / boilerplate code
- ✗ Zero STL containers or memory management
- ✗ No unit tests or CI pipelines

CANDIDATE B — THE EVIDENCE-BACKED BUILDER

“Intermediate C++”



INSPECTION LOG

- ✓ 4 C++ repositories analyzed — 2,840 lines parsed
- ✓ Extensive STL: `std::vector`, `std::unordered_map`, custom iterators
- ✓ CLI system interaction + multi-module build `CMakeLists.txt`
- ✓ Contribution defense: 84% authored by candidate (commit velocity verified)

Candidate B wrote “Intermediate”. Candidate A wrote “Advanced”. Resumes cannot tell the difference. **ProofBridge can.**

Enterprise-Grade, Deterministic, and Self-Hosted

ZERO BLOCKCHAIN · ZERO GAS FEES

LAYER 01

Frontend & Experience

Next.js 14 App Router

TypeScript

Tailwind CSS

Lucide Icons

Framer Motion

CLIENT-SIDE VERIFICATION

Cryptographic signature checks run in-browser via [SubtleCrypto](#) — no server round-trip required.

LAYER 02

Data & Vector Intelligence Core

- ▶ PostgreSQL on Supabase with [Row-Level Security \(RLS\)](#)
- ▶ [pgvector](#) embedding store for role-to-skill semantic clustering
- ▶ Deterministic Role Coverage Algorithm — identical inputs always yield identical scores

ROLE COVERAGE FORMULA

$$\text{Score} = \sum (W_i \times \min(1, S_{lvl} / R_{lvl})) \times \text{Mandatory_Multiplier}$$

LAYER 03

W3C Verifiable Credentials Ledger

Decentralized Identifiers (DIDs) conforming to the W3C Verifiable Credentials Standard v2.0.

CRYPTOGRAPHIC PAYLOAD

Student_DID
+ Skill_ID
+ Faculty_Signature ([Ed25519](#))
+ Timestamp
+ [SHA-256 Merkle Proof](#)

- ✓ Zero recurring cloud gas fees
- ✓ Zero blockchain latency
- ✓ Verifiable offline by any standard HR system

Macro Analytics for Higher Education Governance

MCA / B.TECH COHORT 2026 - 620 STUDENTS

CURRICULUM GAP MATRIX

SKILL DOMAIN	DEMAND vs. READINESS	GAP	STATUS
React & Frontend demand 81% · ready 72%		9%	• STRONG
SQL & Databases demand 78% · ready 74%		4%	• HEALTHY
Git & Team Collab demand 86% · ready 58%		28%	• WARNING
Docker & Containers demand 67% · ready 22%		45%	• CRITICAL
Cloud Deployment demand 71% · ready 27%		44%	• CRITICAL

AI ACADEMIC ACTION
RECOMMENDATION

⚠ DEAN ALERT - INTERVENTION REQUIRED

Inject a **2-week intensive Docker + CI/CD sprint** into **Semester 5 · Lab 304**.

45%

gap closed across the cohort

178

additional students qualified
for hiring drives

BLIND MATCHING PROTOCOL – recruiters review candidates strictly through cryptographic score cards before reveal, destroying brand and demographic bias.

Zero Recurring Friction, Maximum Institutional ROI

\$0

RECURRING INFRA COST

\$40,000/yr

LMS SAVINGS PER CAMPUS

85%

SCREENING TIME REDUCTION

1-click

NAAC / NBA AUDIT EXPORT

PILLAR 01

Economic Feasibility

- ▶ Built on open-source PostgreSQL, local compute parsers, and edge serverless APIs.
- ▶ Eliminates predatory per-seat LMS subscription models.

\$40,000 / yr saved per campus

PILLAR 02

NAAC & NBA Accreditation Automation

- ▶ Criterion 1 — Curricular Aspects
- ▶ Criterion 2 — Teaching-Learning & Evaluation
- ▶ NBA Outcome-Based Education course outcome attainment proofs

Panic reporting → audit-ready telemetry

PILLAR 03

Recruiter ROI

42h -85% **6h**
BEFORE AFTER

Initial technical screening cycle compressed per hire, with no loss of signal.

100% pre-vetted before round 1

Engineering Team Outliers & What's Next

Akshar Miyani

LEAD ARCHITECT

Next.js · AST Parsers · Graph Matching Engines

Lubhanshi Mathur

FULL-STACK & UI/UX

Design Systems · Student Passport Dashboard

Lakshita Sharma

INTELLIGENCE PIPELINES

Role Genome Modeling · Skill Gap Radars

Mohammad Faizan

SYSTEMS & CRYPTOGRAPHY

Verifiable Credentials · DB Schema & RLS

3-PHASE EXECUTION ROADMAP

PHASE 1 • DELIVERED

Hackathon Final Build

Native SaaS platform, GitHub AST parser, Role Genome matching, Dean's Curriculum Radar, live PostgreSQL backend.

PHASE 2 Q3 2026

Federation & Sandbox

Multi-university federation across the Manipal network, plus an automated WASM-based code execution sandbox.

PHASE 3 Q4 2026

National & ATS Integration

Direct integration with the national skill registry and enterprise ATS platforms — Workday, Greenhouse.

EXPLORE THE PROTOTYPE

LIVE APPLICATION

<http://localhost:3000>

GITHUB REPOSITORY

github.com/miyaniakshar1234/IIC3.0-Final

“Turning student capability into verifiable currency.”